

Replacement memo: corrected AR(1) Section 5.3

SVP simulation audit

15 September 2026

Decision

Section 5.3 is replaced by the calibrated four-method AR(1) power study described below. The original `SVP_NEW.pdf` remains untouched because no editable source was available. The pre-refresh AR(1) outputs are preserved in `simulations/other_simus/power_ar1_legacy_20260915/`.

Study design

We simulate independent realizations of the piecewise-constant mean model

$$Y_t = \mu_t + E_t, \quad E_t = \rho E_{t-1} + \eta_t, \quad \rho = 0.8,$$

with $n = 600$ observations and unit stationary marginal noise variance. The innovations are Gaussian with variance $1 - \rho^2$, and the first residual is drawn from the stationary distribution. The four signal patterns are `none`, `up`, `updown`, and `rand1`. Jump sizes range from 1 to 5 in increments of 0.2, with 100 replications per configuration.

Methods and calibration

The competitors are PELT AR1 approximate, PELT inflated, DeCAFS AR1, and SVP AR1Focus. The approximate PELT method applies PELT to the transformed series

$$Z_t = Y_{t+1} - \rho Y_t,$$

and maps each transformed endpoint j to endpoint $j + 1$ in the original series. It is approximate because ordinary transformed-data least squares does not retain the complete AR(1) likelihood boundary correction.

DeCAFS uses the pure AR(1) state-space model with `sdEta = 0`, `sdNu = sqrt(1-rho^2)`, and `phi = rho`. Its multiplier c_D in $\beta = c_D \log(n)$ is selected as the smallest candidate attaining corrected null F1 at least 0.99. The approximate PELT multiplier and the AR1Focus threshold are selected to match the selected DeCAFS null F1. AR1Focus uses `subtests = "right"` only. The inflated PELT baseline retains the fixed penalty $3 \log(n)(1 + \rho)/(1 - \rho)$. The approximate-Pelt multiplier is calibrated alongside DeCAFS and AR1Focus.

Calibration and independent validation each use 1,000 stationary AR(1) null series, deterministic seeds, and Wilson confidence intervals. The complete candidate table and selected values are stored with the study outputs.

Table 1: Validation null F1 for the selected AR(1) methods.

method	candidate	null_f1	null_f1_lower	null_f1_upper
DeCAFS AR1	7.75	0.985	0.9754	0.9909
PELT AR1 approximate	0.80	0.989	0.9804	0.9938
SVP AR1Focus / right	1.85	0.993	0.9856	0.9966
PELT inflated	3.00	0.999	0.9944	0.9998

Table 2: Mean refreshed-minus-legacy metric change by pattern and method.

pattern	method	F1	Precision	Recall	Correct	MSE	Segments
none	DeCAFS AR1	0.0381	0.0381	0.0381	0.0381	-0.0016	-0.0381
none	PELT AR1 approximate	-0.0090	-0.0090	-0.0090	-0.0090	0.0016	0.0105
none	PELT inflated	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
none	SVP AR1Focus	-0.0033	-0.0033	-0.0033	-0.0033	0.0004	0.0033
rand1	DeCAFS AR1	-0.0797	-0.0704	-0.0731	-0.0019	0.1612	-0.5376
rand1	PELT AR1 approximate	0.1037	0.0427	0.1635	-0.0200	-0.3443	2.7629
rand1	PELT inflated	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
rand1	SVP AR1Focus	0.1177	0.0932	0.1168	0.0143	-0.2183	0.9067
up	DeCAFS AR1	-0.0477	0.0039	-0.0689	-0.1290	0.2085	-0.5633
up	PELT AR1 approximate	-0.0039	-0.1705	0.1678	-0.0252	-0.6099	4.0681
up	PELT inflated	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
up	SVP AR1Focus	0.1367	0.0652	0.1605	0.2305	-0.5985	1.0838
updown	DeCAFS AR1	-0.1578	-0.1365	-0.1599	-0.1438	0.3199	-1.1590
updown	PELT AR1 approximate	0.2168	0.1307	0.3476	-0.0210	-0.7302	5.4981
updown	PELT inflated	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
updown	SVP AR1Focus	0.2416	0.2467	0.2394	0.2395	-0.4443	1.7462

Metrics

For every method, the mandatory terminal endpoint n is excluded before one-to-one matching within a tolerance of five observations. We report precision, recall, F1, the probability of recovering the correct number of changepoints, signal MSE, the number of fitted segments, and localization error. The same metric implementation is used by the Gaussian and AR(1) studies.

Before/after comparison

The detailed paired comparison is stored in `simulations/z_AR1/paired_comparison.csv`; it compares the refreshed results with the archived pre-refresh study by pattern and jump size.